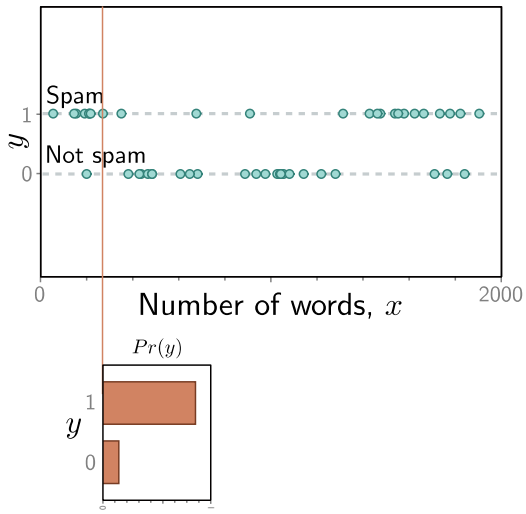
Agenda

- Earlier
  - Typical workflow of a data scientist (Four steps)
  - Estimator and estimate
  - Three methods to get an estimate
  - Point estimate
  - Likelihood vs probability
- Today
  - MLE for normal
  - Linear regression



2

Context and workflow

3

Problem Definition

- Given some data samples $D = \{x_1, x_2, \ldots, x_I\}$
- Given some intended distribution $\Pr(\mathbf{x}_{1\ldots I}|\boldsymbol{\theta})$
  - Bernoulli: $\theta = p$ (probability of success)
  - Poisson: $\theta = \lambda$ (rate parameter)
  - Uniform: θ is a vector of the upper and lower bounds
  - Normal: θ is a vector of μ and σ^2
- Find the value of the parameter that "optimally" fits the distribution assuming $\mathrm{X}_1, \mathrm{X}_2, \ldots, \mathrm{X}_n$ are mutually independent (given θ)

4

Optimal Fitting parameters to model

Three methods

5

Maximum Likelihood (MLE)

- As the name suggests: find the parameters under which the data $x_{1\ldots I}$ are most likely
- When viewed as a function of Θ $p(D|\Theta)$ is called the likelihood function and it is not necessarily a probability
- When viewed as a function of X_i, the conditional probability $p(D|\Theta)$ is a probability distribution that sums to one

$$\hat{\boldsymbol{\theta}} = \arg\max_{\boldsymbol{\theta}} \Pr(\mathbf{x}_{1\ldots I}|\boldsymbol{\theta})$$
$$= \arg\max_{\boldsymbol{\theta}} \left[\prod_{i=1}^{I} \Pr(\mathbf{x}_i|\boldsymbol{\theta})\right]$$

6

MLE Derivation

7

Fitting normal distribution: ML

- As the name suggests we find the parameters under which the data $x_{1\ldots I}$ is most likely

$$\hat{\boldsymbol{\theta}} = \arg\max_{\boldsymbol{\theta}}\Big[\Pr(\mathbf{x}_{1\ldots I} \mid \boldsymbol{\theta})\Big]$$
$$= \arg\max_{\theta}\left[\prod_{i=1}^{I}\Pr(\mathbf{x}_i \mid \theta)\right]$$

Likelihood given by pdf

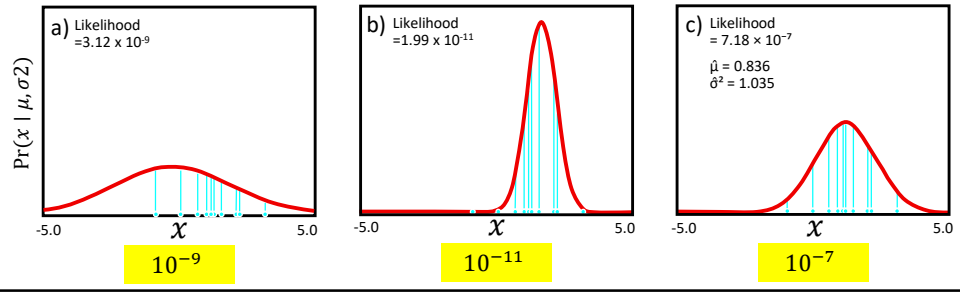
$$\Pr(x \mid \mu, \sigma^2) = \mathrm{Norm}_x[\mu, \sigma^2] = \frac{1}{\sqrt{2\pi\sigma^2}}\exp\left[-0.5\frac{(x-\mu)^2}{\sigma^2}\right]$$

8

Fitting normal distribution: ML

$$\Pr(x_{1\ldots I} \mid \mu, \sigma^2) = \prod_{i=1}^{I}\Pr(x_i \mid \mu, \sigma^2)$$
$$= \prod_{i=1}^{I}\mathrm{Norm}_{x_i}[\mu, \sigma^2]$$
$$= \frac{1}{(2\pi\sigma^2)^{I/2}}\exp\left[-0.5\sum_{i=1}^{I}\frac{(x_i-\mu)^2}{\sigma^2}\right]$$

Ten data points



9

Fitting normal distribution: ML

Algebraically: $\hat{\mu}, \hat{\sigma}^2 = \arg\max_{\mu,\sigma^2} \left[\Pr(x_{1\ldots I} \mid \mu, \sigma^2) \right]$

where: $\Pr(x \mid \mu, \sigma^2) = \text{Norm}_x[\mu, \sigma^2]$

$$\Pr(x_{1\ldots I} \mid \mu, \sigma^2) = \prod_{i=1}^{I} \text{Norm}_{x_i}[\mu, \sigma^2] \quad = \frac{1}{(2\pi\sigma^2)^{I/2}} \exp\left[-0.5 \sum_{i=1}^{I} \frac{(x_i - \mu)^2}{\sigma^2} \right]$$

or alternatively, we can maximize the **logarithm**

$$\begin{aligned}\hat{\mu}, \hat{\sigma}^2 &= \arg\max_{\mu,\sigma^2} \left[\sum_{i=1}^{I} \log\left[\text{Norm}_{x_i}[\mu, \sigma^2] \right] \right] \\ &= \arg\max_{\mu,\sigma^2} \left[-0.5I\log[2\pi] - 0.5I\log\sigma^2 - 0.5\sum_{i=1}^{I} \frac{(x_i - \mu)^2}{\sigma^2} \right]\end{aligned}$$

10

Fitting normal distribution

$$\begin{aligned}\hat{\mu}, \hat{\sigma}^2 &= \arg\max_{\mu,\sigma^2} \left[\sum_{i=1}^{I} \log\left[\text{Norm}_{x_i}[\mu, \sigma^2] \right] \right] \\ &= \arg\max_{\mu,\sigma^2} \left[-0.5I\log(2\pi) - 0.5I\log\sigma^2 - 0.5\sum_{i=1}^{I} \frac{(x_i - \mu)^2}{\sigma^2} \right]\end{aligned}$$

$$\begin{aligned}\frac{\partial L}{\partial \mu} &= \sum_{i=1}^{I} \frac{(x_i - \mu)}{\sigma^2} \\ &= \frac{\sum_{i=1}^{I} x_i}{\sigma^2} - \frac{I\mu}{\sigma^2} = 0\end{aligned}$$

Solution: $\hat{\mu} = \frac{\sum_{i=1}^{I} x_i}{I}$

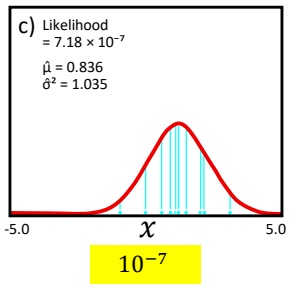
11

Fitting normal distribution: ML

Maximum likelihood solution:

$$\hat{\mu} = \frac{\sum_{i=1}^{I} x_i}{I}$$

$$\hat{\sigma}^2 = \sum_{i=1}^{I} \frac{(x_i - \hat{\mu})^2}{I}$$



Looks familiar?!

12

MLE

- Bernoulli: $\hat{p}_{\mathrm{ML}} = \frac{1}{n}\sum_{i=1}^{n} x_i = \bar{x}$
- Poisson: $\hat{\lambda}_{\mathrm{ML}} = \frac{1}{n}\sum_{i=1}^{n} x_i = \bar{x}$
- Uniform: $\hat{v} = X_{(1)} = \min_i X_i\,, \qquad \hat{\theta} = X_{(n)} = \max_i X_i$

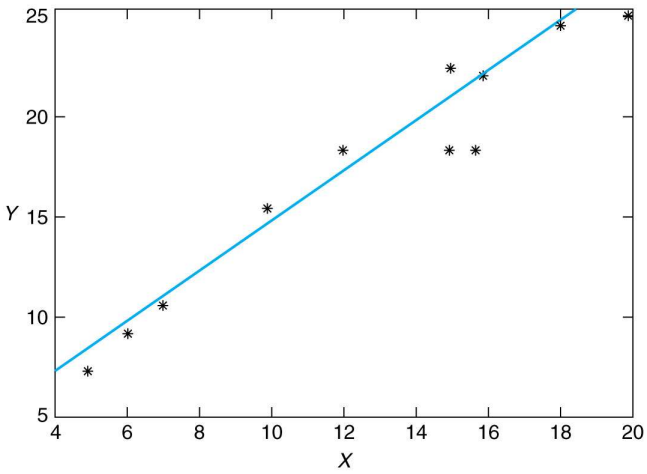
Bernoulli has 1 parameter "p"
uniform has 2 parameters the lower and upper bounds

13

Linear Regression

14

Linear Regression

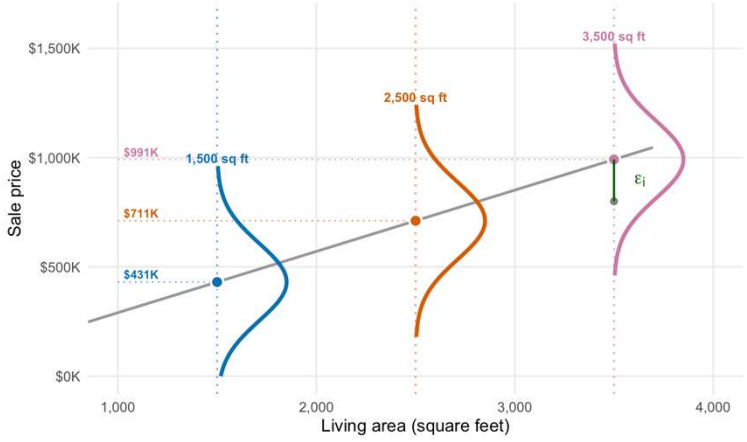


Most common case:
k=1

15

15

Linear Regression



16